

Representation of Words

Clear Notes - Part 1: Foundations

1 Why Represent Words?



- Machine learning algorithms need numbers, not raw words.
- Text is converted into features that models can compare and learn from.
- Granularity moves from documents to sentences to phrases to words.



2 Bag of Words and TF-IDF

Two classical views

Bag of Words: count vocabulary terms in each document.

TF-IDF: weight terms higher when they are frequent in one document but rare overall.

apple	banana	car	dog	fish
2	1	1	0	0
0.81	0.57	0.25	0	0

Each dimension corresponds to a vocabulary term.

3 One-Hot Encoding

- Each word gets a vector with one 1 and all other values 0.
- Useful for representing individual vocabulary items.
- Vocabulary size directly becomes vector length.

Vocabulary: [a, cat, sat, on, the]

a	1	0	0	0	0
cat	0	1	0	0	0
sat	0	0	1	0	0
the	0	0	0	0	1

Example: 'a cat sat on the' becomes a sequence of one-hot vectors.

4 Limits of Count Vectors



- Word order is ignored by BoW and TF-IDF.
- Different words are equally far apart in one-hot space.
- Vectors are sparse and high-dimensional.

Order problem

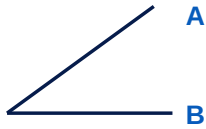
"apple banana" → [1, 1, 0, 0]

"banana apple" → [1, 1, 0, 0]

Representation of Words

Clear Notes - Part 2: Embeddings

1 Cosine Similarity



$$\cos(A,B) = \frac{(A \text{ dot } B)}{(\|A\| \|B\|)}$$

For one-hot vectors, different words usually have dot product 0, so cosine similarity is 0.

2 Dense Word Vectors

- Instead of huge sparse one-hot vectors, learn dense low-dimensional vectors.
- Rows of the embedding matrix become word embeddings.
- Similar usage gives nearby vectors.

Sparse one-hot

word



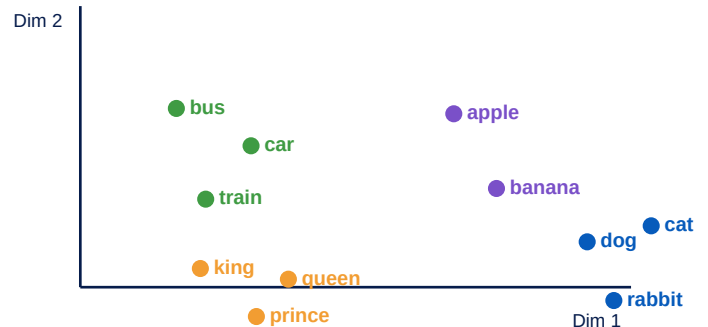
Dense embedding



3 Distributional Hypothesis

Words in similar contexts
tend to have similar meanings.

'I traveled by bus' and 'I traveled by car' teach the model that bus and car are close.



4 Quick Comparison

Feature	Integer Coding	One-Hot	Learned Embeddings
False order problem	Yes	No	No
Compactness	Yes	No	Yes
Distance meaning	No	No	Yes
Semantic similarity	No	No	Yes

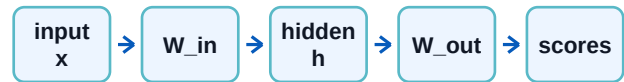
Representation of Words

Clear Notes - Part 3: Word2Vec

1 Word2Vec Idea



- Word2Vec learns embeddings from nearby words.
- Two common setups: CBOW and Skip-gram.
- The learned rows of W_{in} are used as word embeddings.



V = vocabulary size, N = embedding dimension.

2 CBOW and Skip-gram



- CBOW uses surrounding context words to predict the center word.
- Skip-gram uses the center word to predict surrounding context words.

Example sentence



Context with window = 2: [a, cat, on, the] $\rightarrow$ sat

CBOW: context $\rightarrow$ center
Skip-gram: center $\rightarrow$ context

3 CBOW Input Vector

Vocabulary: [a, cat, sat, on, the]

a	1	0	0	0	0
cat	0	1	0	0	0
on	0	0	0	1	0
the	0	0	0	0	1

Average context one-hot:

$x = [0.25, 0.25, 0, 0.25, 0.25]$

This average vector is the numeric input to the CBOW model.

4 Embedding Matrix

Word	Dim 1	Dim 2
a	0.2	0.1
cat	0.9	0.8
sat	0.7	0.6
on	0.1	0.3
the	0.6	0.2

- W_{in} has shape $V \times N$, here 5×2 .
- $E(\text{word})$ means the row vector for that word.
- $h = \text{average}(E(a), E(\text{cat}), E(\text{on}), E(\text{the}))$.

h represents the context, not a single word.

Representation of Words

Clear Notes - Part 4: CBOW Training

1 Tiny Corpus

Small corpus

S1: the cat sat on mat
S2: the dog sat on mat
S3: the cat ate fish

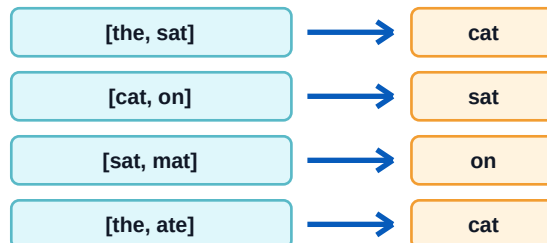
Vocabulary ($V = 8$)

the	cat	sat	on
mat	dog	ate	fish

2 Training Pairs

- Slide a window over each sentence.
- Use nearby words as context and the middle word as the target.
- Every window becomes one training example.

CBOW examples



3 Forward Pass

Example input: [cat, on] → sat

cat	0	1	0	0	0	0	0	0
on	0	0	0	1	0	0	0	0

$$x = 0.5 \text{ cat} + 0.5 \text{ on} = [0, 0.5, 0, 0.5, 0, 0, 0, 0]$$

Hidden vector:

$$h = 0.5 E(\text{cat}) + 0.5 E(\text{on})$$

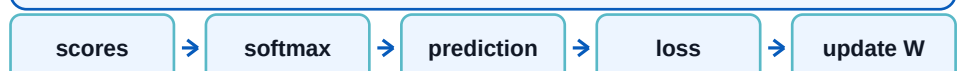
4 Scores and Softmax

Scores before softmax

word	score
the	0.13
cat	0.57
sat	1.30
on	0.72

$$\text{score}(\text{word}) = h \cdot E_{\text{out}}(\text{word})$$

Softmax converts scores into probabilities. Training increases the probability of the true center word, here 'sat'.



Representation of Words

Clear Notes - Part 5: Classical vs Neural

1 What Good Vectors Need



- The model needs better numbers than arbitrary IDs.
- Useful vectors should be addable, comparable, and learnable.
- Distances should carry meaning whenever possible.

"the cat sat on the mat" -> vectors a model can use

2 Classical ML Representation

raw text



tokens



dictionary



counts



sparse vector

D1: the cat sat on the mat | D2: the dog sat on the log | D3:
the cat saw the dog

Classical ML usually uses BoW or TF-IDF: simple, sparse, and strong for many tasks.

3 Neural Representations

- Word2Vec learns one vector per word from context.
- RNN/LSTM models learn from word order over a sequence.
- Contextual models can give the same word different vectors in different sentences.

bank near a river

bank for money

● finance sense

● river sense

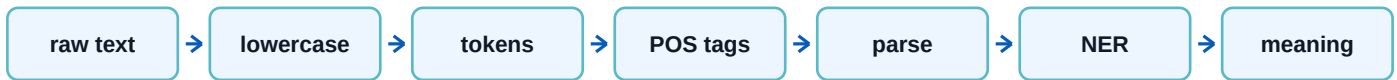
4 Revision Table

Method	Dense	Order	Meaning	Best use
BoW	No	No	Weak	quick baseline
TF-IDF	No	No	term weight	classical ML
Word2Vec	Yes	local context	strong	similarity
Contextual	Yes	Yes	changes by context	modern NLP

Representation of Words

Clear Notes - Part 6: NLP Tagging

1 Text Processing Pipeline



- Tokenization splits text into words, punctuation, and sentences.
- Lowercasing and punctuation handling make inputs consistent.
- Later steps add grammar and entity information.

2 POS Tagging

Tag	Meaning	Examples	Role
DT	Determiner	the, a	points to noun
NN	Noun	dog, cat	thing/person
VB	Verb	run, eat	action
JJ	Adjective	big, red	description

The/DT dog/NN
runs/VB fast/JJ

3 Sequence Model Intuition

- A sequence tagger chooses the best tag path, not each tag independently.
- Transition probability: how likely one tag follows another.
- Emission probability: how likely a word appears with a tag.

$$\text{score} = p(\text{tag} \mid \text{prev tag}) \times p(\text{word} \mid \text{tag})$$

Used in HMM/Viterbi style POS tagging.

4 Mini Probability View

Piece	Example	Meaning	Value
Transition	$p(\text{NN} \mid \text{DT})$	noun after determiner	high
Emission	$p(\text{dog} \mid \text{NN})$	dog as noun	high
Transition	$p(\text{VB} \mid \text{NN})$	verb after noun	medium

Main idea: numbers become useful when they preserve context, order, and meaning.